# Galileo's comparison as two-path interference

A coherent, endpoint-matched comparison of Newton's impulsive chord with Galileo's constant-force parabola has action difference

$$K_\tau := \Delta S = \frac{F^2\tau^3}{24m} = \frac{F}{4v}A_{\text{inertial,fall}} = \frac{\tau\Delta E}{12}.$$

At supplied reduced phase parameter $\hbar > 0$, its optimal equal-prior error against the zero-phase reference, using $n$ independent controlled passes and an arbitrary joint measurement, is exactly

$$p_{\text{err}}^{(n)} = \frac{1}{2}\left(1 - \sqrt{1 - \cos^{2n}\left(\frac{K_\tau}{2\hbar}\right)}\right).$$

For fixed $n, \hbar$, refinement sends this error to $1/2$. At fixed $\tau > 0$, the coherent error oscillates as $\hbar \downarrow 0$ and has no limit. A specified averaging of the inverse phase parameter gives a genuine noncommutation theorem for coherence; a specified tunable comparison gives one for distinguishability (Theorem 3). These prescriptions make precise the extra operation needed in the proposed classical-first limit.

The resulting action threshold is conditional on the phase rule, the Born rule and the resources of the comparison. It re-expresses the polygon-lift phase bound through the halved functional. Theorems C and E of the fifth-postulate note constrain different records with the same supplied quantum action constant. Positivity of that constant remains an independent physical obligation. All results below are proved under their displayed hypotheses; the scale-selection step remains open.

The symbol $\hbar$ is the supplied reduced quantum phase constant, with physical Planck action $h_P = 2\pi\hbar$. The time mesh and instrument resolution remain separate. Varying or averaging inverse values in the limiting theorems below is a formal family-of-models prescription, not an experiment that varies nature's constant or an automatic record-resolution law.

The comparison uses controlled unitaries and product copies; Proposition 2 is not used in the limiting Theorem 3. Its relative averaging prescription is part of the hypothesis. Model checks are internal, not independent review.

## 1. The Galileo area and the two histories

Let $m, F, v, \tau > 0$, with downward coordinate $y$ and common horizontal motion $x = vt$. The initial tangent and falling curve are

$$y_I(t) = 0, \qquad y_F(t) = \frac{Ft^2}{2m}, \qquad A_{\text{inertial,fall}} = \frac{vF\tau^3}{6m}, \qquad \Delta E = \frac{F^2\tau^2}{2m}. \tag{1}$$

Thus $(3F/v)A_{\text{inertial,fall}} = \tau\Delta E$. The common-endpoint chord is $y_C(t) = F\tau t/(2m)$. For the transverse Lagrangian $L = m\dot{y}^2/2 + Fy$, direct integration gives

$$S[y_I] = 0, \qquad S[y_F] = \frac{F^2\tau^3}{3m}, \qquad S[y_C] = \frac{3F^2\tau^3}{8m}, \qquad S[y_C] - S[y_F] = K_\tau. \tag{2}$$

The horizontal action cancels in every difference. Equation (2) also exposes the endpoint issue: the bare tangent and parabola finish at different heights. Adding $d\chi(y,t)/dt$ to $L$ changes their action difference by their unequal final $\chi$ values. Their measurable relative phase therefore requires endpoint matching and specified control phases. The conversion coefficient for $K_\tau$ is $F/(4v)$; the area conversion $(F/v)A_{\text{inertial,fall}}$ equals $4K_\tau$.

Here is the exact matching used throughout. Supply canonical quantum kinematics $[y, p] = i\hbar$ in the Schrödinger representation, and set

$$U_\tau = \exp\left[-\frac{i\tau}{\hbar}\left(\frac{p^2}{2m} - Fy\right)\right], \qquad Q_\tau = e^{iF\tau y/(2\hbar)} e^{-i\tau p^2/(2m\hbar)} e^{iF\tau y/(2\hbar)}.$$

**Lemma 1 (matched comparison).** With these scalar-potential conventions,

$$Q_\tau = e^{iK_\tau/\hbar} U_\tau. \tag{3}$$

*Proof.* The kernel of $Q_\tau$ has the free normalization and phase $L_\tau(x,y) = m(y-x)^2/(2\tau) + F\tau(x+y)/2$. The classical solution joining $x$ to $y$ is $q_*(t) = x + (y-x)t/\tau + F(t^2 - \tau t)/(2m)$. Substitution into the action gives $L_\tau - K_\tau$. The fluctuation action about $q_*$ is exactly $\int m\dot{\xi}^2/2$, with zero endpoint fluctuations, hence the same free Gaussian normalization. This proves (3), equivalently the exact quadratic kernel identity in the insertion note, §§2–3 (passage). □

The impulses in $Q_\tau$ reproduce the chord and match the parabola's final momentum after the last half-kick. Equation (3) holds for every initial body state. A branch-dependent scalar potential would add its own phase, so its absence or calibration is part of the experiment. The polygon-lift note, Theorem 1 (full-read) proves the corresponding identity for every partition.

For a literal tangent arm, the explicit reunion in the Galileo note, §§3–4 (full-read) uses signal forces $F, -F, F$ over durations $\tau, 2\tau, \tau$ against a free reference. Its closed action difference is $\Delta S_{\text{loop}} = -2F^2\tau^3/(3m) = -16K_\tau$. Every phase formula below applies with that signed difference when that four-cell experiment is chosen. The coefficient belongs to the reunion protocol; the original area (1) remains the geometric anchor.

For fixed endpoints the constant-force action has one stationary path. The chord is an off-shell path for the continuous-force action and a classical history for the impulsive control. Thus (3) compares two controlled histories, each with an exact quadratic path integral. Treating the tangent and parabola as two stationary points of one fixed-endpoint constant-force functional would change the variational problem.

## 2. What the halved functional measures

Use $\hbar$ for reduced phase parameter and $\eta \in [0, 1/2)$ for a target decision error. Identification with quantum mechanics means $\hbar = |\hbar|$, with ordinary Planck constant $h_P = 2\pi|\hbar|$.

**Proposition 2 (two contributions and their remainder).** Suppose a halved functional, with its normalization and observable fixed, has two retained contributions

$$\mathcal{A}_\hbar = a_0 e^{iS_0/\hbar + i\mu_0} + a_1 e^{iS_1/\hbar + i\mu_1} + R_\hbar, \quad a_j \geq 0, \quad |R_\hbar| \leq r.$$

Put $\phi = (S_1 - S_0)/\hbar$ and $\beta = \mu_1 - \mu_0$. Then

$$|\mathcal{A}_\hbar|^2 = a_0^2 + a_1^2 + 2a_0a_1\cos(\phi + \beta) + E, \qquad |E| \leq 2(a_0 + a_1)r + r^2. \tag{4}$$

*Proof.* Expand the square, bound the leading amplitude by $a_0 + a_1$, and bound its cross term with $R_\hbar$ by $2(a_0 + a_1)r$. □

For two nondegenerate stationary points at a fixed partition, $a_j = |O(z_j)|/\sqrt{|\det S''(z_j)|}$ and $\mu_j$ contains the observable phase and Hessian signature. Stationary phase supplies $r \leq C_\pi\hbar$ for sufficiently small $\hbar$, with cutoffs and smoothness as in round 1, §1 (full-read). Equation (4) retains that partition-dependent constant; uniform control as the partition changes needs a separate estimate. For the exactly matched quadratic propagators in (3), the relative phase has zero remainder and zero relative Gaussian signature.

At a fixed phase, a large $|S_1 - S_0|/\hbar$ still leaves the cross term in (4) at full amplitude. Averaging, selecting a branch, or retaining an orthogonal branch record supplies the missing diagonalization. Equal-action contributions survive phase averaging, as round 1's pair of quartic saddles demonstrates.

An operational record requires a preparation and measurement rule in addition to (4). For example, tag the arms by normalized pointer states $|r_0\rangle, |r_1\rangle$, and put $\gamma = \langle r_0|r_1\rangle$. Recombining the arms gives the cross term $2a_0a_1\,\mathrm{Re}(\gamma e^{i(\phi+\beta)})$ and visibility $V = 2a_0a_1|\gamma|/(a_0^2 + a_1^2)$. For balanced arms and pure tags, optimal inference of the arm from the pointer has

$$D_{\mathrm{path}} = \sqrt{1 - |\gamma|^2}, \qquad p_{\mathrm{path}} = \frac{1 - D_{\mathrm{path}}}{2}, \qquad V^2 + D_{\mathrm{path}}^2 = 1. \tag{5}$$

The eigenvalues of the difference of the two rank-one tag states are $\pm\sqrt{1 - |\gamma|^2}$, proving (5) by the binary measurement optimization used below. This is the pure balanced case of established interferometric duality; see Schwindt, Kwiat and Englert 1999 (abstract). Path identification depends on the tags, while phase detection depends on coherent recombination. Orthogonal tags identify the arm even at $\Delta S = 0$ and remove its interference. A bare halved functional supplies neither those tags nor their physical cost.

## 3. Exact phase record, error target and resources

Prepare a balanced control qubit and a common arbitrary body density operator $\rho$. Under hypothesis $H_0$, apply $U_\tau$ in both arms; under $H_1$, apply $U_\tau$ in arm 0 and $Q_\tau$ in arm 1. Supply coherent control with calibrated relative scalar phases, no extra environmental tags, the Born rule, and equal hypothesis priors. Equation (3) factorizes the output into the same body state under both hypotheses and the control states

$$|+\rangle = \frac{|0\rangle + |1\rangle}{\sqrt{2}}, \qquad |+_\phi\rangle = \frac{|0\rangle + e^{i\phi}|1\rangle}{\sqrt{2}}, \qquad \phi = K_\tau/\hbar. \tag{6}$$

A recombiner of phase $\alpha$ has exact port probabilities $P_\pm(\alpha|\phi) = [1 \pm \cos(\phi - \alpha)]/2$. Scanning $\alpha$ reveals the fringe shift; its visibility remains one for every $\phi$. The optimal distinction of the two hypotheses is

$$D_n = \sqrt{1 - \cos^{2n}(\phi/2)}, \qquad p_{\mathrm{err}}^{(n)} = \frac{1 - D_n}{2}. \tag{7}$$

*Proof.* For normalized vectors with overlap $c$, the difference of their rank-one density operators has eigenvalues $\pm\sqrt{1 - |c|^2}$. For any equal-prior binary test $0 \le M \le I$, the success probability is $[1 + \mathrm{tr}\, M(\rho_1 - \rho_0)]/2$; the positive eigenspace attains the maximum. Here $|\langle +|+_\phi\rangle| = |\cos(\phi/2)|$ and $n$ independent copies raise the overlap to its $n$th power. This proves (7) including attainability. □

Thus (7) detects a history-dependent phase against a reference. The uncontrolled body channels of $Q_\tau$ and $U_\tau$ coincide for every $\rho$, giving error 1/2 even with arbitrarily many copies. The coherence resource in (6) is essential to exposing the scalar.

For the original unclosed free/falling pair, the final displacement and impulse themselves supply a further signal. A pure body's overlap then contains the expectation of a Weyl displacement in its prepared state, so its magnitude depends on that preparation. For each $\tau > 0$ one can choose a wavefunction with position support narrower than the nonzero interaction-picture displacement $F\tau^2/(2m)$; its translated support is disjoint and the two outputs are orthogonal. Shrinking that support costs increasing momentum spread. Thus the phase-only theorem applies to the matched, coherently controlled record (6), with its specified resources; an unrestricted detector of the raw endpoint separation requires a separate preparation or disturbance budget.

Write $\delta = \mathrm{dist}(\phi, 2\pi\mathbb{Z}) \in [0, \pi]$. For a fixed positive integer $n$, (7) reaches error at most $\eta$ iff

$$\delta \ge d_{n,\eta}, \qquad d_{n,\eta} = 2\arccos\left([4\eta(1-\eta)]^{1/(2n)}\right). \tag{8}$$

On the first lobe $0 \le K_\tau/\hbar \le \pi$, this is equivalent to

$$K_\tau \geq \hbar d_{n,\eta}, \qquad \tau \geq \left(\frac{24m\hbar d_{n,\eta}}{F^2}\right)^{1/3}, \qquad A_{\text{inertial,fall}} \geq \frac{4v\hbar}{F} d_{n,\eta}. \tag{9}$$

For $n = 1$, $d_{1,\eta} = 2\arcsin(1-2\eta)$, exactly the polygon-lift record bound, §4 (passage). For $0 < \eta < 1/2$, $d_{n,\eta} \to 0$ as $n \to \infty$; for $\eta = 0$ it equals $\pi$ at every finite $n$. With $k$ coherent passes replace $\phi$ by $k\phi$. These resource choices change the threshold.

The dependence on errors is explicit. If actual hypothesis states $\widetilde{\rho}_i$ satisfy $\frac{1}{2}\|\widetilde{\rho}_i - \rho_i\|_1 \leq e_i$, the optimum error changes by at most $(e_0 + e_1)/2$, by the triangle inequality in (7). A common phase calibration error $u$ on each of $n$ independent signal copies gives distance at most $\sqrt{1-\cos^{2n}(u/2)} \leq \sqrt{n}\,|u|/2$, and hence changes the optimal error by at most $\sqrt{n}\,|u|/4$ when the reference is exact. An action uncertainty $|\delta S|$ contributes $|u| = |\delta S|/\hbar$. These bounds state how accurately the control must be known to use (9).

## 4. The two orders of limits

Put $C = F^2/(24m) > 0$, so $K_\tau = C\tau^3$.

**Theorem 3 (refinement, classical resolution and the missing prescription).**

(a) For the fixed coherent protocol (6), with fixed finite $n$,

$$D_n \leq \min\left(1, \frac{\sqrt{n}\,C\tau^3}{2\hbar}\right), \qquad \lim_{\tau\downarrow 0} p_{\text{err}}^{(n)} = \frac{1}{2} \quad (\hbar > 0). \tag{10}$$

At every fixed $\tau > 0$, $\lim_{\hbar\downarrow 0} D_n$ fails to exist: its liminf is zero and limsup is one. Consequently the coherent classical-first iterated limit is undefined.

(b) Fix $b \in (0,1)$ and average the inverse phase parameter by replacing $\phi$ with $u\phi$, uniformly over $u \in [1-b, 1+b]$, forgetting $u$. For this explicitly averaged two-arm state, its normalized coherence is

$$\Gamma_b(\tau,\hbar) = \frac{1}{2b}\int_{1-b}^{1+b} e^{iuK_\tau/\hbar}\,du = e^{i\phi}\,\text{sinc}(b\phi), \qquad |\Gamma_b| \leq \min\left(1, \frac{\hbar}{bK_\tau}\right). \tag{11}$$

With $\text{sinc}(0) = 1$, the iterated limits are

$$\lim_{\hbar\downarrow 0}\lim_{\tau\downarrow 0}\Gamma_b = 1, \qquad \lim_{\tau\downarrow 0}\lim_{\hbar\downarrow 0}\Gamma_b = 0. \tag{12}$$

(Refinement after a Fable review, 2026-09-27.) The non-commuting limits (12) depend on averaging over a *relative* window of the inverse resolution. With an absolute window of half-width $w_0$ in $1/\hbar$ the average is $e^{i\phi}\,\text{sinc}(K_\tau w_0)$, independent of $\hbar$, and both iterated limits equal one. Theorem 3(b) therefore states a property of the relative-averaging prescription, which is the prescription (12) uses.

For fixed weights and signature difference in (4), this prescription leaves a diagonal-intensity error bounded by $2a_0a_1\min(1, \hbar/(b|\Delta S|))$ in addition to the averaged remainder bound of (4), with bound $2a_0a_1$ when $\Delta S = 0$. If the stationary-phase remainder obeys $r \leq C_\pi\hbar$ uniformly for small resolution, averaging uses resolutions $\hbar/u$ and one may take $r \leq C_\pi\hbar/(1-b)$.

(c) Alternatively, assume a calibrated comparison can be attenuated to any action $rK_\tau$, $0 \leq r \leq 1$, under both hypotheses, with no additional relative phase. For example, use duration $r^{1/3}\tau$ in (6), keeping $F, m$ fixed. Let $D_{\text{tun}}$ be the best single-copy distance over these choices. Then

$$D_{\mathrm{tun}}(\tau,\hbar)=\sin\left(\min\left\{\frac{K_\tau}{2\hbar},\frac{\pi}{2}\right\}\right), \tag{13}$$

$$\lim_{\hbar\downarrow 0}\lim_{\tau\downarrow 0} D_{\mathrm{tun}}=0, \qquad \lim_{\tau\downarrow 0}\lim_{\hbar\downarrow 0} D_{\mathrm{tun}}=1. \tag{14}$$

*Proof.* The inequality $1-(1-x)^n\le nx$ for $x\in[0,1]$, with $x=\sin^2(\phi/2)$, proves (10). For fixed $K_\tau>0$, the sequences $\hbar_j=K_\tau/(2\pi j)$ and $\hbar'_j=K_\tau/((2j+1)\pi)$ give respectively $D_n=0$ and $D_n=1$. Integration gives (11); dominated continuity at $\phi=0$ and the displayed reciprocal bound at $\phi\to\infty$ prove (12). In (c), maximize $|\sin(r\phi/2)|$ over $[0,1]$. The first maximum is reached at $r=\pi/\phi$ when $\phi\ge\pi$; otherwise $r=1$ maximizes it. This proves (13)–(14). □

Part (b) formalizes the loss of off-diagonal terms needed to recover the two selected critical-point weights. Its averaged balanced state approaches $I/2$ classically first, and $|+\rangle\langle+|$ when refined first. The trace distance between these two limiting states is $1/2$; dephasing by itself supplies no accessible which-path tag. Part (c) instead supplies an actual discrimination experiment with errors tending respectively to $1/2$ and zero. Its extra tunable control is a physical hypothesis; changing the output recombiner alone would leave (7) unchanged. A classical branch-selection prescription can also retain distinct histories at every positive cell length, but their paths converge together as that length vanishes. A limiting Dirac measure alone expresses no retained record of the shrinking difference.

These conclusions sharpen the proposed classical-first statement. For any fixed $b$ the condition $b|\Delta S|/\hbar\gg 1$ controls the averaged cross term, with the explicit error in (11). For coherent discrimination the exact condition is (8), with its phase aliases. The two conditions express different measurements. Along a joint limit the dimensionless ratio $C\tau^3/\hbar$ governs both: tending to zero erases the phase signal; a finite ratio gives the corresponding signal, including the aliases in (8); a diverging ratio requires the prescription in (b) or (c) for a definite limit.

For a partition of fixed total duration $T$, the matched polygon action is $K_\pi=C\sum_j\tau_j^3\le CT|\pi|^2$. Its coherent distance from the exact curve is at most $\sqrt{n}\,CT|\pi|^2/(2\hbar)$. After $k$ uniform halvings a cell's action falls by $8^{-k}$ and the whole polygon defect by $4^{-k}$. Hence the whole fixed-duration comparison converges as well. Resolving ever smaller cells can require increasing resources: for a fixed small-error target the small-phase independent-copy scaling is $n$ of order $(\hbar/K_\tau)^2$, whereas coherent repetition uses $k$ of order $\hbar/K_\tau$. Such budgets were held fixed in (10).

## 5. Relation to the disturbance and concentration floors

The fifth-postulate note, Theorems C, E and §7 (passage) separates observable algebra, states and records. With $\hbar=|\hbar|$, the three bounds here have the following scopes.

| Bound | Quantity constrained | Extra hypotheses beyond Newtonian geometry |
|---|---|---|
| Phase record (9), one copy | $K_\tau\ge 2\|\hbar\|\arcsin(1-2\eta)$ on the first lobe | Coherent controlled comparison, calibrated reunion, Born rule and one pass |
| Theorem C(b) | $(s/8)\sum_j\Delta D_j+(J/2)\sum_j\Delta X_j\ge\|\hbar\|\arcsin(1-2\eta)$ | Regular quantum joint states, unitary instrument, test valid for every initial body state; the theorem's interpolating-state suprema |

| Bound | Quantity constrained | Extra hypotheses beyond Newtonian geometry |
|---|---|---|
| Theorem E | $\lambda_0(ab/\lvert\hbar\rvert) \geq (1-2\eta)^2$ | Quantum state concentrated in both position and momentum windows with probability at least $1-\eta$ each |

Here $s = F\tau^2/(2m)$, $J = F\tau$, $a, b$ are window half-widths, and $\lambda_0$ is the largest eigenvalue of the time-and-band-limiting operator with the fifth-postulate note's convention. For Theorem E's finite threshold take $0 < \eta < 1/2$; at $\eta = 0$ simultaneous compact concentration is impossible for finite $a, b$. A phase difference of $\pi$ already gives perfect discrimination in (7).

The geometry relates $sJ = 12K_\tau$. Converting Theorem C into a bound on $K_\tau$ additionally needs bounds on the allowed disturbances $\Delta D_j, \Delta X_j$ relative to $J, s$. Converting Theorem E needs a physical identification of the concentration widths $a, b$ with trajectory resolution. Choosing, for example, $a = \alpha s$ and $b = \beta J$ would give $K_\tau \geq |\hbar| c_*(\eta)/(12\alpha\beta)$, with $c_*(\eta) = \lambda_0^{-1}((1-2\eta)^2)$ and the positive dimensionless apertures $\alpha, \beta$ supplied. Geometry alone fixes neither choice.

The phase experiment has a two-dimensional record space. Its Fubini–Study distance from the reference is $\delta/2$, and the equal-prior error target requires that distance to be at least $\arcsin(1-2\eta)$. This is the same statistical-angle step used in Theorem C's disturbance proof. Theorem E uses the overlap norm of two spectral projections in an infinite-dimensional canonical state space. Its concentration function and error dependence carry different information. Their common right-hand scale comes from the supplied canonical phase constant; the equality of the scale alone establishes no equivalence of the three experimental tasks.

There is a stronger exact bridge within the restricted Gaussian mark model. The polygon-lift note, Theorem 7 (passage) uses the same functional $\mathcal{K}_\tau[f] = (2m)^{-1} \iint G_\tau(s,u) f(s) f(u)\, ds\, du$ for the polygon phase and the optimal mark deflection. For constant force, $\mathcal{K}_\tau = K_\tau$; with mark trade-off $\kappa = |\hbar|/2$, its supremal squared deflection is $d^2_{\mathrm{mark}} = 2K_\tau/|\hbar|$. That equality retains the Gaussian, uncorrelated error/recoil and dense-protocol hypotheses. It leaves the general C and E bounds with their own quantities and quantifiers.

Thus the two-path calculation gives the polygon theorem's existing phase bound in the halved-functional representation, plus its explicit limit prescriptions. It adds no independent exclusion of the common zero branch allowed by round 2, §4 (passage). Every $\hbar > 0$ supports (3)–(14); their thresholds scale with that freely supplied value. At zero, the exponential representation is singular, while classical mechanics still supplies ordinary trajectories and, with state completeness, sharp pointer records as in Theorem C(c). A singular coordinate for that classical theory creates no inconsistency in the theory itself.

## 6. Consequence for STATE

Item 2 gains an exact two-path error law on the Galileo cubic action, with the endpoint conversion, phase aliases, resources and limiting prescriptions explicit. Fixed-resolution refinement erases this phase record; classical-first averaging or calibrated attenuation gives the precise contrasting limits. The result locates a conditional record threshold at $K_\tau = \tau\Delta E/12$ for the matched polygon protocol, and at the corresponding calibrated action for a literal inertial reunion. The remaining necessity step is to justify a common positive phase/readout scale independently of the quantum premises and a fixed experimental budget. Round 3 supplies the representation and its exact obstruction, while that physical scale-selection problem remains open.